

LEBANON, NH, USA

WMO: 726116

Lat: 43.626N

Lon: 72.305W

Elev: 598

StdP: 14.38

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 94765

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -9.1	(c) -2.7	(d) -17.4	(e) 2.2	(f) -6.0	(g) -12.0	(h) 2.9	(i) 0.1	(j) 19.9	(k) 28.5	(l) 18.1	(m) 25.1	(n) 2.1	(o) 350	(p) 0.515

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.1	89.2	71.3	86.1	69.8	83.2	68.2	74.3	84.3	72.7	81.7	71.1	79.2	8.3	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
71.6	119.3	79.2	70.0	112.7	77.3	68.2	105.9	75.9	38.2	84.2	36.6	81.5	35.2	79.2	80.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.9	15.9	13.1	-18.3	94.1	6.1	2.6	-22.7	96.0	-26.3	97.6	-29.7	99.0	-34.1	100.9
			-18.6	77.6	6.0	1.6	-23.0	78.8	-26.5	79.7	-29.8	80.6	-34.2	81.8
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	46.2	19.8	22.4	31.7	44.5	56.6	64.9	70.2	68.4	61.3	48.7	37.7	26.5		
	DBStd	19.52	12.03	10.70	9.87	8.61	7.75	6.64	5.31	5.45	7.14	8.12	8.21	10.20		
	HDD50	3747	935	772	571	209	23	0	0	0	3	127	379	728		
	HDD65	7321	1400	1192	1033	615	280	82	14	28	156	509	820	1192		
	CDD50	2361	0	0	4	45	228	448	627	572	342	85	9	1		
	CDD65	459	0	0	0	2	20	80	176	135	44	2	0	0		
	CDH74	4574	0	0	5	71	352	865	1704	1177	380	20	0	0		
	CDH80	1415	0	0	1	19	101	268	587	335	104	0	0	0		
(14) Wind	WSAvg	4.5	4.6	4.9	5.5	5.6	4.8	4.3	3.9	3.6	3.7	4.2	4.5	4.3		
(15) Precipitation	PrecAvg	35.6	2.4	2.1	2.4	2.7	3.4	3.0	3.2	3.3	3.2	3.1	3.4	3.0		
	PrecMax	48.9	5.9	6.6	4.2	5.1	8.9	6.2	7.2	10.0	6.3	7.0	6.8	7.1		
	PrecMin	23.5	0.4	0.1	0.4	1.1	1.1	1.0	0.6	1.3	0.6	0.3	1.4	1.0		
	PrecStd	6.6	1.6	1.1	0.9	0.9	1.8	1.5	1.7	1.8	1.6	1.9	1.6	1.5		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	51.0	53.1	67.2	82.0	88.2	92.0	93.2	91.1	88.7	76.9	66.0	55.3		
		MCWB	47.0	44.6	52.1	61.2	65.7	72.8	74.7	72.9	71.8	63.6	57.7	50.7		
	2%	DB	43.7	46.0	56.7	73.4	82.3	87.1	89.2	87.0	82.7	71.3	61.2	48.4		
		MCWB	39.0	38.8	46.8	56.0	63.6	70.2	72.0	70.7	68.6	61.4	54.6	44.4		
	5%	DB	39.1	40.8	50.6	67.5	77.6	82.7	86.3	83.8	78.3	66.3	55.4	44.1		
		MCWB	35.1	35.7	41.8	53.1	61.2	67.0	70.5	69.4	66.7	58.6	49.3	40.3		
	10%	DB	35.9	36.8	45.8	61.5	72.6	79.2	83.0	81.0	74.1	62.3	51.4	39.8		
		MCWB	32.8	32.7	39.4	48.9	58.6	65.1	68.9	67.8	64.4	55.7	46.0	36.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	48.3	46.3	55.2	63.6	69.9	75.5	77.1	75.6	73.7	67.1	60.2	51.4		
		MCDB	50.9	51.5	63.3	74.8	81.8	86.0	89.2	85.5	84.0	73.1	63.9	54.7		
	2%	WB	39.9	39.7	47.6	59.2	66.7	72.9	74.7	73.4	71.2	63.2	55.6	45.5		
		MCDB	43.2	44.5	54.5	70.7	77.5	82.7	85.0	82.5	78.9	68.5	59.9	48.2		
	5%	WB	35.7	36.3	43.0	54.5	64.0	70.6	72.9	71.8	69.0	59.8	50.7	40.8		
		MCDB	38.6	40.2	49.8	64.2	73.3	78.7	82.0	80.0	75.4	65.3	55.1	43.8		
	10%	WB	33.0	33.5	39.5	50.3	61.2	68.1	71.1	70.1	66.4	56.6	46.3	36.9		
		MCDB	35.4	36.2	45.0	60.3	69.9	75.4	79.4	77.3	72.1	61.2	49.9	39.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	18.9	20.8	20.4	23.0	24.1	22.8	23.1	23.0	22.7	20.3	17.5	16.0		
		MCDBR	19.5	22.7	27.6	35.6	33.0	28.1	26.6	25.7	26.7	26.5	24.7	19.0		
	5% WB	MCWBR	16.4	17.2	18.1	19.5	16.3	13.5	12.0	12.2	14.3	16.6	18.2	15.9		
		MCWBR	18.3	20.9	24.8	30.5	26.7	23.5	23.1	21.6	22.0	22.9	21.5	18.0		
(40) Clear-Sky Solar Irradiance	taub		0.284	0.288	0.319	0.356	0.384	0.429	0.445	0.420	0.370	0.335	0.308	0.281		
	taud		2.385	2.347	2.390	2.390	2.359	2.246	2.205	2.264	2.391	2.501	2.522	2.474		
	Ebn at Noon		269	289	292	288	282	268	262	265	271	269	260	261		
	Edh at Noon		24	30	33	36	38	43	44	40	33	26	22	21		
(44) All-Sky Solar Radiation	RadAvg		500	780	1147	1414	1589	1712	1776	1609	1303	801	532	383		
	RadStd		46	71	75	168	186	156	126	87	90	85	60	39		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+1.08	N/A	N/A	N/A	N/A	-0.88	N/A	N/A	+234	N/A
(47) Regional (59 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+153	+58

Nomenclature: See separate page